Ch#22 Pg # 652

Problem 1)

$E_{net} = E_1 + E_2$

$= \frac{kq}{r^2} + \frac{kq}{r^2}$

Given Data

$q_1 = -2 \times 10^{-7} C$

$q_2 = 2 \times 10^{-7} C$

$x_1 = 6\ cm = 0.06 m$

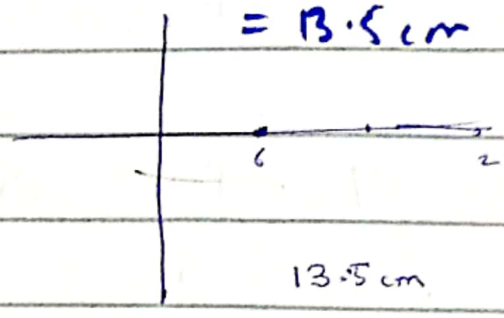
$x_2 = 0.21 m$

$E_{net} = ?$

Mid Distance

$x = \frac{x_1 + x_2}{2} = \frac{6+21}{2}$

$= 13.5 cm$



Sol.

$\cancel{E_1 = \frac{kq}{r^2} \Rightarrow \frac{8.99 \times 10^9 \times 2 \times 10^{-1}}{3.6 \times 10^{-3}} \Rightarrow 4.99 \times 10^{9-7+3}}$

Midway point

$x = 13.5\ cm \Rightarrow 0.135 m$

1)

$E = \frac{kq}{r^2} \Rightarrow \frac{8.99 \times 10^9 \times 2 \times 10^{-7}}{(0.135 - 0.06)^2}$

$\Rightarrow \frac{17.98 \times 10^{9-7}}{5.625 \times 10^{-3}} \Rightarrow 3.19 \times 10^{2+3} \Rightarrow \boxed{3.19 \times 10^5 N/C}$

Same Value in E_2 $E_1 = E_2 = (3.19 \times 10^5)\hat{i}$

$E = E_1 + E_2 \Rightarrow -3.19 \times 10^5 - 3.19 \times 10^5$

$E = (-6.39 \times 10^5\ N/C)\hat{i}$

5) bw wins

Given Data

$E = 2\ N/C$

$r = 50cm \Rightarrow 0.5m$

$q = ?$

$E = k\frac{q}{r^2}$

$\frac{rE}{k} = q$

Sol

$E = k\frac{q}{r^2}$

Ans

$q = \frac{E \times r^2}{k} \Rightarrow \frac{2 \times 0.5^2}{8.99 \times 10^9} \Rightarrow \boxed{5.6 \times 10^{-11} C}$

Q.7)

Given Data

$q_1 = 10\,nC$, $q_2 = -20\,nC$

$q_3 = +20\,nC$, $q_4 = -10\,nC$

$a = 5cm \Rightarrow 0.03m$

$E_{net} = ?$

Sol

$E_{net} = E_1 + E_2 + E_3 + E_4$

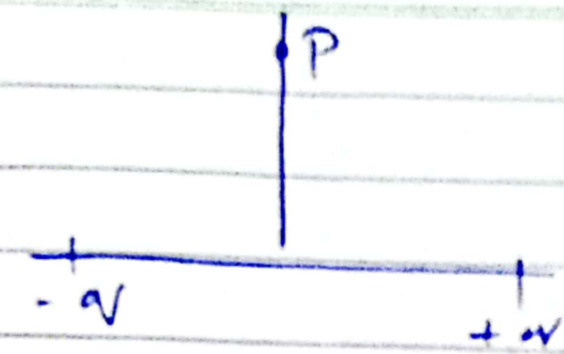
$= \left(\frac{kq}{r^2} + \frac{kq}{r^2} + \frac{kq}{r^2} + \frac{kq}{r^2}\right)\cos 45°$

$= \frac{k}{a^2}(10 - 20 + 20 - 10) \cdot \frac{1}{\sqrt{2}}$

$\boxed{E_{net} \Rightarrow 0}$

$$x = \frac{3+3+4}{2} \Rightarrow 5$$

1) Given Data



$q = -q = 3.20 \times 10^{-19}$

$x = -3\ m$

$x = 3\ m$

$r = 5\ m$

Sol.

$$E = \frac{kq}{r^2} \cos\theta \Rightarrow \frac{8.99 \times 10^9 \times 3.20 \times 10^{-19}}{(5)^2} \cos\frac{3}{5}$$

$$\Rightarrow \frac{28.768}{25} \times 10^{9-19} \, (0.999)$$

$$\boxed{E_{net} \Rightarrow 1.149 \times 10^{-10}\ N/C}$$ Magnitude

b) Net E.F is directed -neugative direction of the x-axis

$$E = \frac{kq}{r^2} \Rightarrow \frac{8.99 \times 10^9 \times 3.2 \times 10^{-19}}{(5)^2}$$

Ch # 22

Q. 18

$$E_{net} = \frac{1}{2\pi r\varepsilon_0} \frac{qdz}{z^4\left(1-\frac{d^2}{4z^2}\right)^2}$$

$$\left(1-\frac{d^2}{4z^2}\right)^{-2} = 1+\frac{2d^2}{4z^2} \qquad (1-x)^{-n} = 1+nx+\frac{n(n-1)}{2!}x^2$$

$$E_{net} = \frac{1}{2\pi r\varepsilon_0} \frac{qd}{z^3}\left(1+\frac{2d^2}{4z^2}\right)$$

$$= \frac{1}{2\pi\varepsilon_0}\frac{qd}{z^3} + \frac{1}{\pi\varepsilon_0}\frac{qd^3}{4z^5}$$

3 4, 5 16

21, 22 23

25, 26, 28, 29

$$\left(1-\frac{d^2}{4z^2}\right)^{-2} \Rightarrow 1+\frac{2d^2}{4z^2} \Rightarrow 1+\frac{d^2}{2z^2}$$